Wassam Hussain

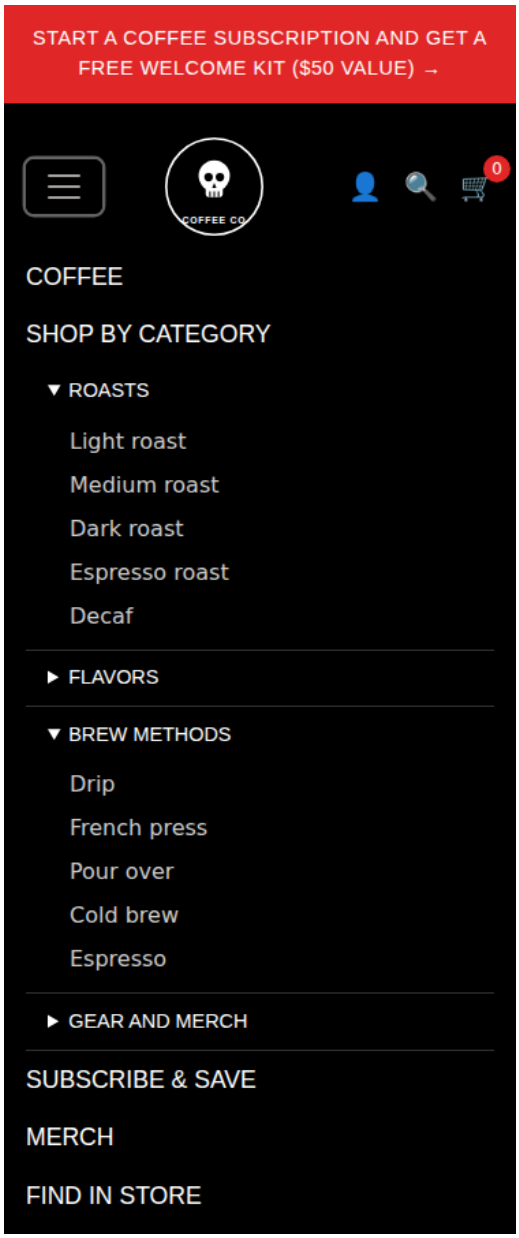
Registration No.: FA24-BCS-131

CSC336 Web Technologies, Lab Mid Term: Upgrade Your E-commerce Navigation, Checkout and Cart
Personal numbers: N = 31, BP = 631px, H = 217 (accent hsl(217, 70%, 45%)), K = 4, max adds per product = 5. Delivery window used in Task 2: 2026-10-29 to 2026-11-29.

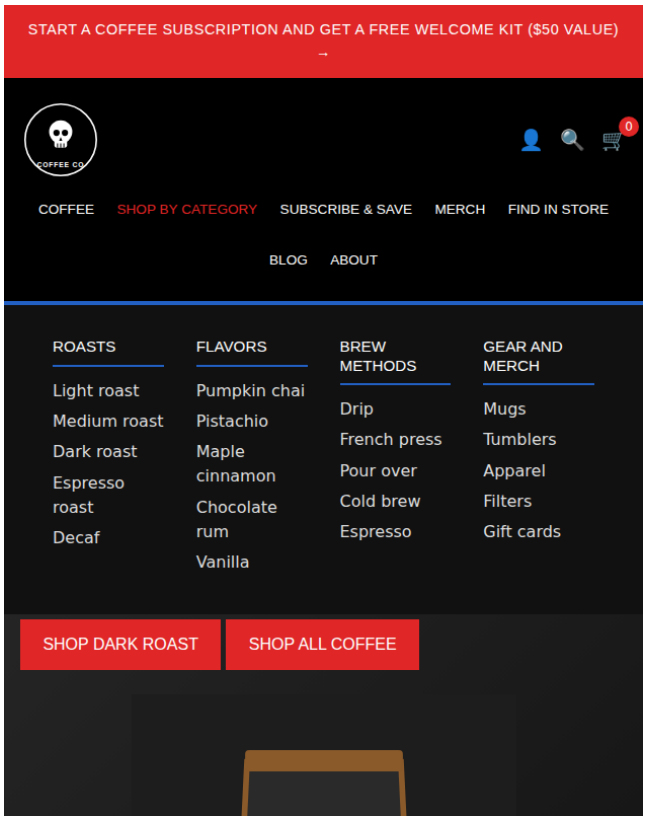
Screenshots

Each task was opened in the browser at 375px, exactly BP (631px) and 1280px.

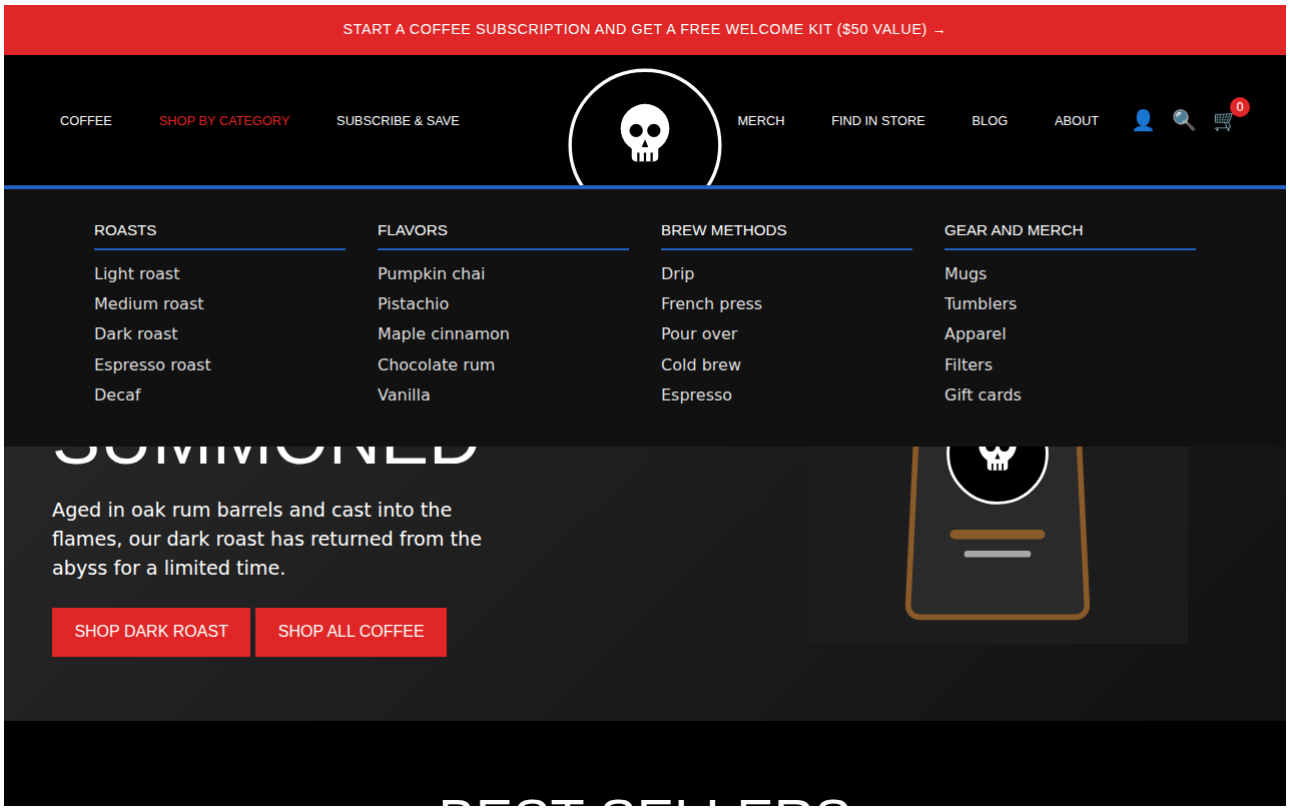
Task 1: Mega-menu



375px: hamburger menu open, accordion with Roasts and Brew methods opened independently

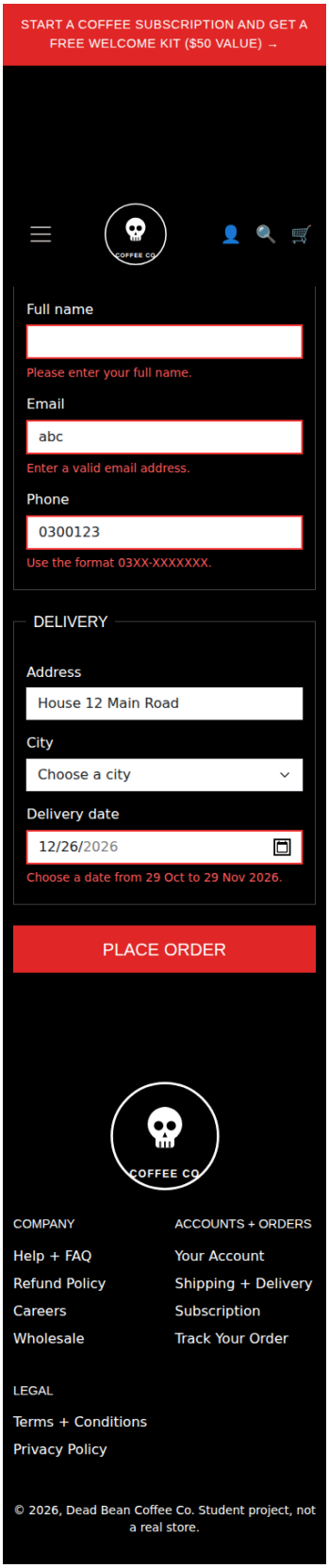


631px (BP): hover on Shop by Category opens the 4-column panel

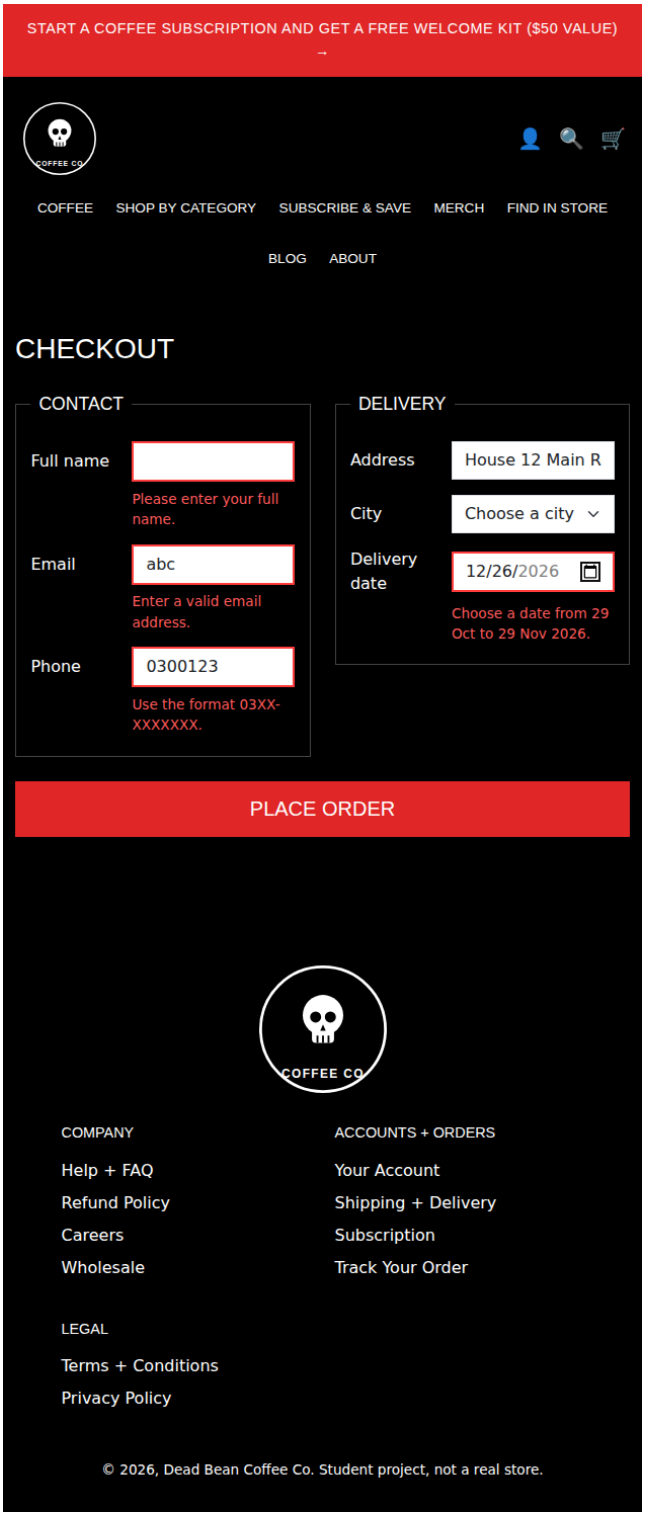


1280px: mega-menu panel open (hover or Tab focus), over the hero, page content not pushed down

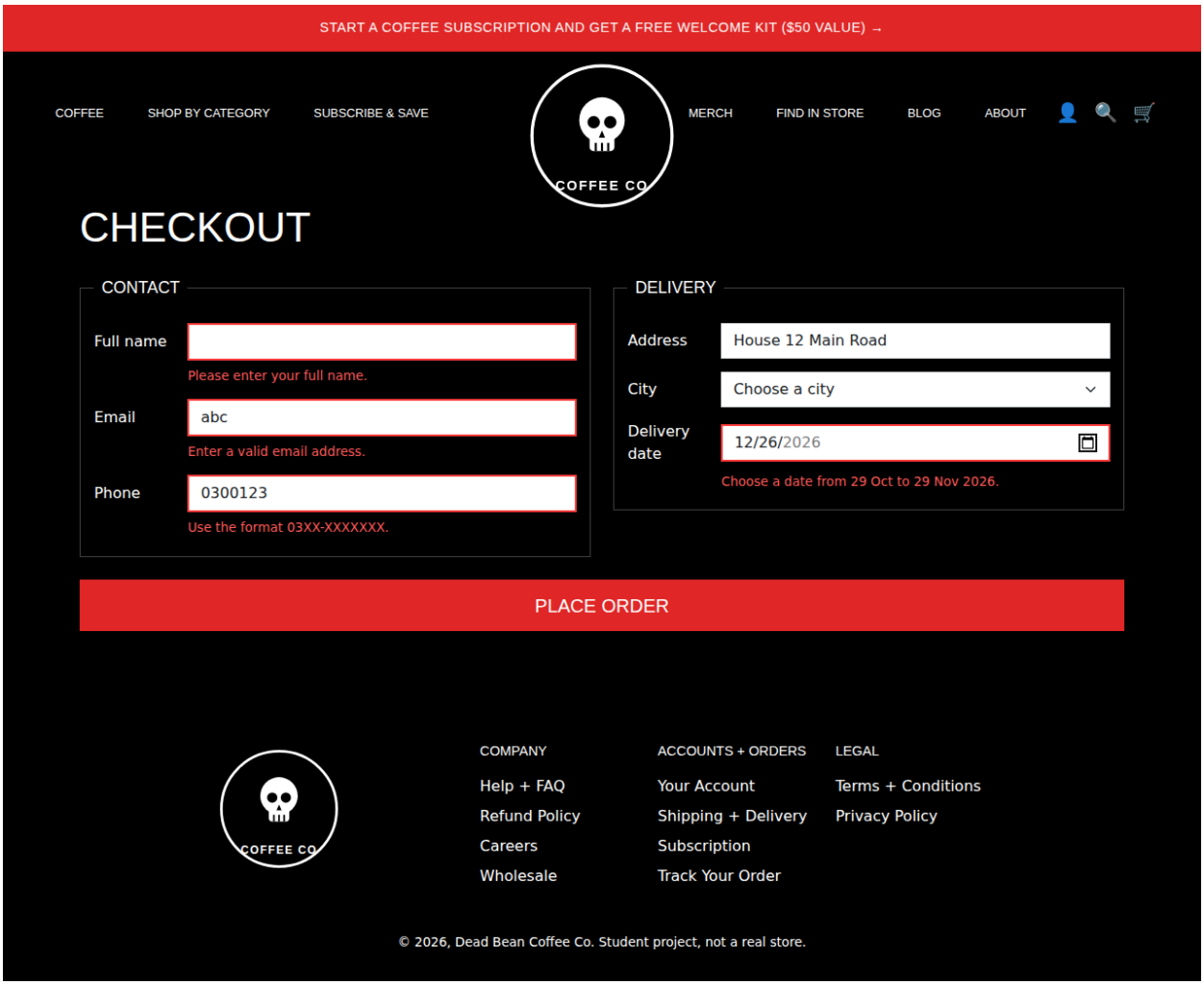
Task 2: Checkout form (HTML5 validation only)



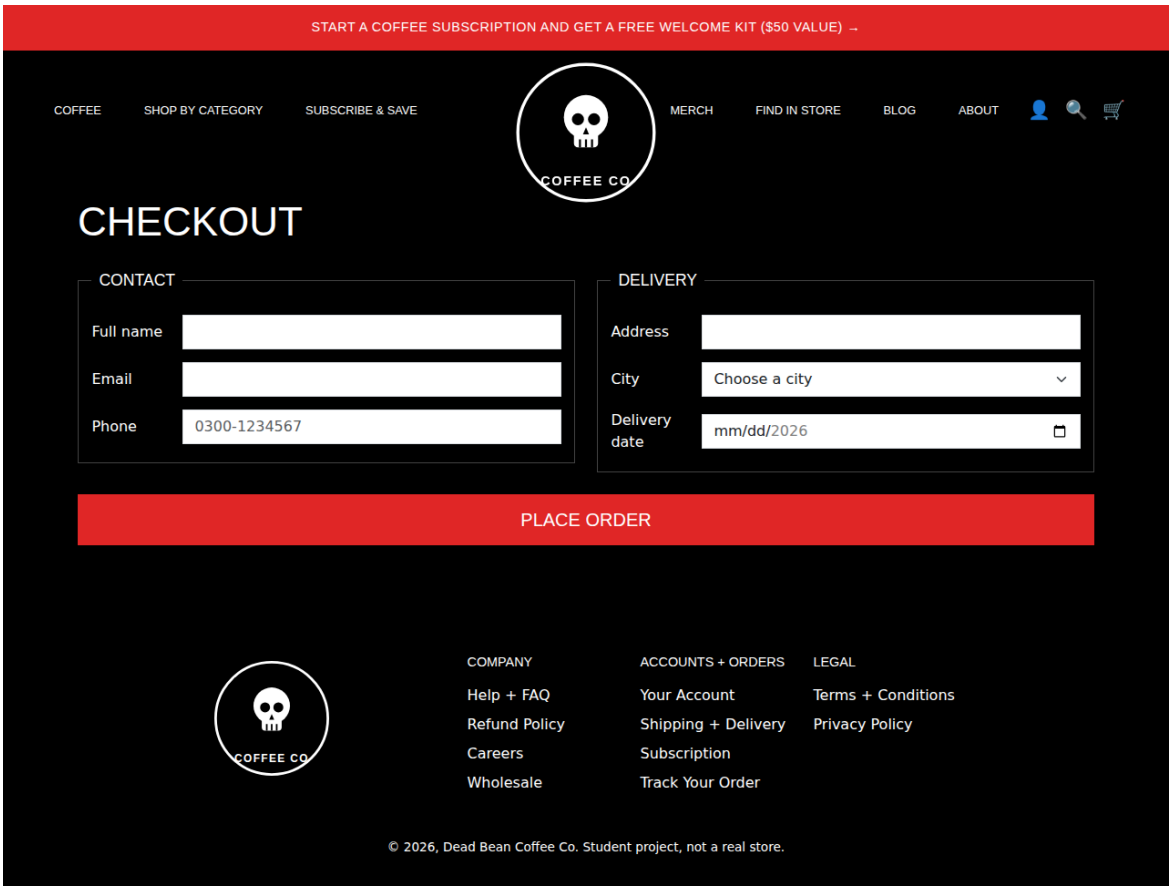
375px: one column, labels above inputs, full-width submit



631px (BP): fieldsets side by side, labels in a left column

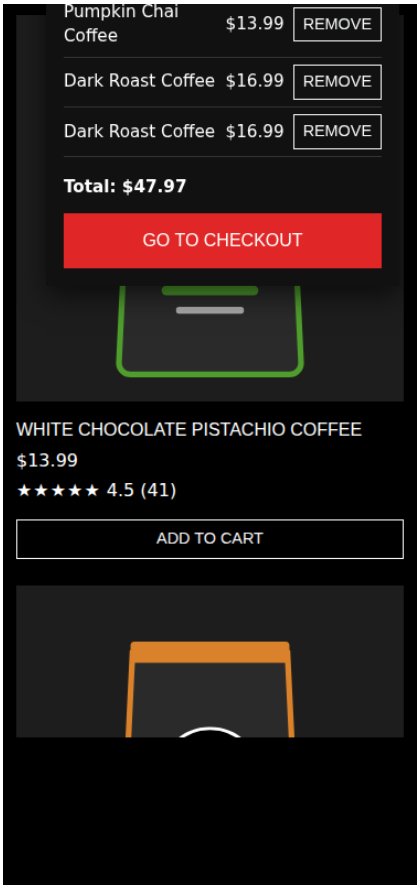


1280px: invalid fields (red border and message) shown only after interaction

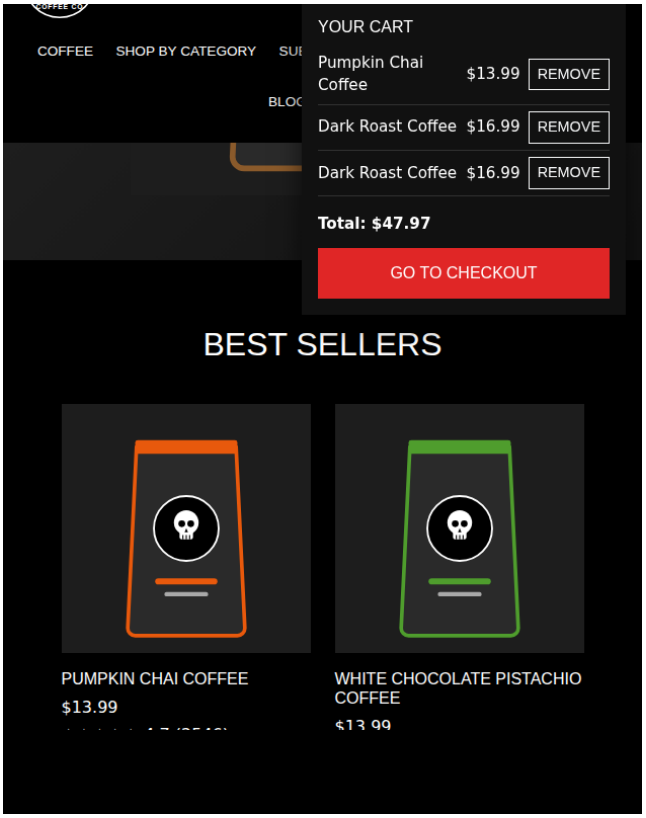


1280px: page on load, no red borders and no messages

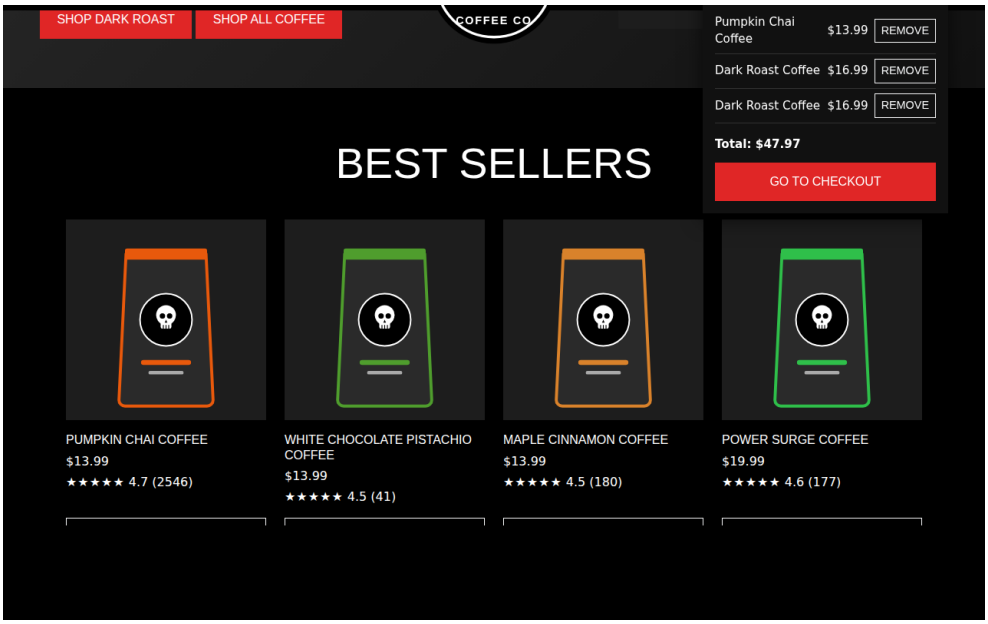
Task 3: Cart badge and mini cart



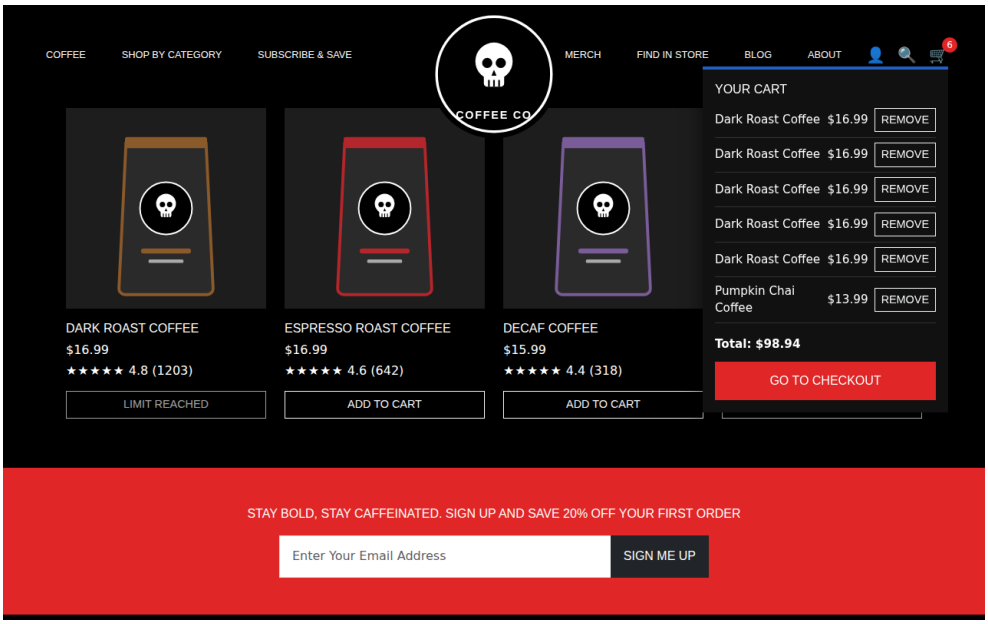
375px: 3 items added, badge shows 3, mini cart open with total



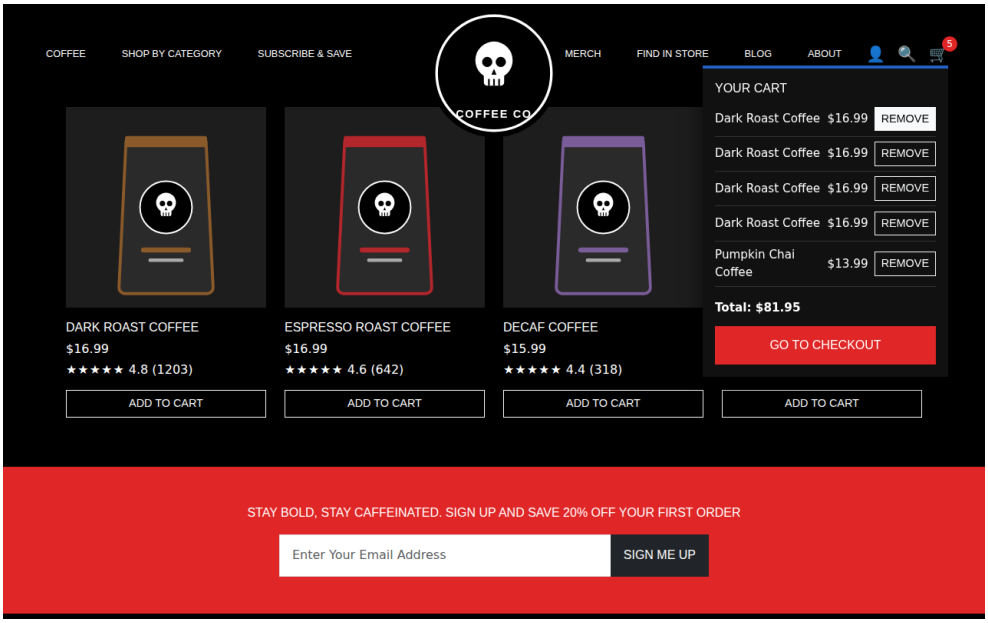
631px (BP): same after using the cart



1280px: badge, mini cart with Remove buttons and total calculated with reduce()



1280px: Dark Roast added 5 times, button shows Limit reached and is disabled



1280px: after pressing Remove, the list, total and badge update and the button can be used again

Code

Task 1: Mega-menu in the navbar

HTML added to the navbar (index.html; the same item is in product.html and checkout.html). In the header I also removed navbar-expand-lg and the order-lg-1 / order-lg-2 classes, because the hamburger now switches at 631px (done in CSS), and I moved the logo-gap item after Subscribe & Save.

index.html (navbar item)

```
<!-- TASK 1 START: mega-menu item -->
            <li class="nav-item mega">
              <a class="nav-link" href="#">Shop by Category</a>

              <!-- from BP and wider: panel with K = 4 columns -->
              <div class="mega-panel">
                <div class="mega-columns">
              <div class="mega-col">
                <h3>Roasts</h3>
                <ul>
                  <li><a href="#">Light roast</a></li>
                  <li><a href="#">Medium roast</a></li>
                  <li><a href="#">Dark roast</a></li>
                  <li><a href="#">Espresso roast</a></li>
                  <li><a href="#">Decaf</a></li>
                </ul>
              </div>
              <div class="mega-col">
                <h3>Flavors</h3>
                <ul>
                  <li><a href="#">Pumpkin chai</a></li>
                  <li><a href="#">Pistachio</a></li>
                  <li><a href="#">Maple cinnamon</a></li>
                  <li><a href="#">Chocolate rum</a></li>
                  <li><a href="#">Vanilla</a></li>
                </ul>
              </div>
              <div class="mega-col">
                <h3>Brew methods</h3>
                <ul>
                  <li><a href="#">Drip</a></li>
                  <li><a href="#">French press</a></li>
                  <li><a href="#">Pour over</a></li>
                  <li><a href="#">Cold brew</a></li>
                  <li><a href="#">Espresso</a></li>
                </ul>
              </div>
              <div class="mega-col">
                <h3>Gear and merch</h3>
                <ul>
                  <li><a href="#">Mugs</a></li>
                  <li><a href="#">Tumblers</a></li>
                  <li><a href="#">Apparel</a></li>
                  <li><a href="#">Filters</a></li>
                  <li><a href="#">Gift cards</a></li>
                </ul>
              </div>
                </div>
              </div>

              <!-- below BP: the same 4 groups as an accordion -->
              <div class="mega-accordion">
              <details>
                <summary>Roasts</summary>
                <ul>
                  <li><a href="#">Light roast</a></li>
                  <li><a href="#">Medium roast</a></li>
                  <li><a href="#">Dark roast</a></li>
                  <li><a href="#">Espresso roast</a></li>
                  <li><a href="#">Decaf</a></li>
                </ul>
              </details>
              <details>
                <summary>Flavors</summary>
                <ul>
                  <li><a href="#">Pumpkin chai</a></li>
                  <li><a href="#">Pistachio</a></li>
                  <li><a href="#">Maple cinnamon</a></li>
                  <li><a href="#">Chocolate rum</a></li>
                  <li><a href="#">Vanilla</a></li>
```

```
              </ul>
            </details>
            <details>
              <summary>Brew methods</summary>
              <ul>
                <li><a href="#">Drip</a></li>
                <li><a href="#">French press</a></li>
                <li><a href="#">Pour over</a></li>
                <li><a href="#">Cold brew</a></li>
                <li><a href="#">Espresso</a></li>
              </ul>
            </details>
            <details>
              <summary>Gear and merch</summary>
              <ul>
                <li><a href="#">Mugs</a></li>
                <li><a href="#">Tumblers</a></li>
                <li><a href="#">Apparel</a></li>
                <li><a href="#">Filters</a></li>
                <li><a href="#">Gift cards</a></li>
              </ul>
            </details>
            </div>
          </li>
          <!-- TASK 1 END -->
```

styles.css (Task 1 section)

```
/* ===== TASK 1: navbar breakpoint and mega-menu =====
   BP = 600 + 31 = 631px, K = 4 columns, accent hue H = 217 */
:root {
  --accent: hsl(217, 70%, 45%);
}

/* Below BP the desktop panel does not exist; from BP up it opens on hover or focus (see below) */
.mega-panel {
  display: none;
}

/* Below BP: accordion made with <details> inside the hamburger menu */
.mega-accordion details {
  border-bottom: 1px solid #333;
}
.mega-accordion summary {
  cursor: pointer;
  padding: 0.6rem 1rem;
  font-family: "Archivo Black", "Archivo", sans-serif;
  font-size: 0.9rem;
  text-transform: uppercase;
}
.mega-accordion ul {
  list-style: none;
  margin: 0;
  padding: 0 0 0.5rem 2rem;
}
.mega-accordion a {
  display: block;
  padding: 0.25rem 0;
  color: #ccc;
  text-decoration: none;
}
.mega-accordion a:hover {
  color: #fff;
  text-decoration: underline;
}

/* From BP and wider: links always visible (no hamburger) and the mega panel works */
@media (min-width: 631px) {
  .navbar-toggler {
    display: none;
  }
  .navbar-collapse {
    display: block !important; /* beats Bootstrap's .collapse:not(.show) */
    flex-basis: 100%;
    order: 3; /* links sit on their own row under the logo and icons */
  }
  .nav-main {
    display: flex;
    flex-direction: row;
    flex-wrap: wrap;
    justify-content: center;
  }
  .nav-main .nav-link {
    font-size: 0.85rem;
```

```
    padding: 0.9rem 0.7rem;
  }

  .mega-accordion {
    display: none;
  }
  .mega-panel {
    display: block;
    visibility: hidden; /* hidden, not display:none, so it can fade and close with a short delay */
    opacity: 0;
    transition: opacity 0.15s, visibility 0s linear 0.3s; /* 0.3s delay on closing */
    position: absolute; /* does not push the page down; anchored to .navbar (position: relative) */
    top: 100%;
    left: 0;
    right: 0;
    z-index: 20; /* inside the sticky header (z-index 1030), so above grid, cards and hero */
    background: #111;
    border-top: 4px solid var(--accent);
    padding: 2rem 3rem;
  }
  /* invisible bridge: keeps the panel open while the pointer crosses the gap */
  .mega-panel::before {
    content: "";
    position: absolute;
    left: 0;
    right: 0;
    bottom: 100%;
    height: 1.5rem;
  }
  .mega:hover .mega-panel,
  .mega:focus-within .mega-panel {
    visibility: visible;
    opacity: 1;
    transition-delay: 0s; /* open immediately */
  }
  .mega-columns {
    display: grid;
    grid-template-columns: repeat(4, 1fr); /* K = 4 */
    gap: 2rem;
    max-width: 1100px;
    margin: 0 auto;
  }
  .mega-col h3 {
    font-size: 0.95rem;
    padding-bottom: 0.5rem;
    margin-bottom: 0.75rem;
    border-bottom: 2px solid var(--accent);
  }
  .mega-col ul {
    list-style: none;
    margin: 0;
    padding: 0;
  }
  .mega-col li {
    margin-bottom: 0.4rem;
  }
  .mega-col a {
    color: #ddd;
    text-decoration: none;
  }
  .mega-col a:hover {
    color: #fff;
    text-decoration: underline;
  }
}

/* From 1200px: big centred logo overlapping the hero, as in Assignment 2 */
@media (min-width: 1200px) {
  .logo {
    position: absolute;
    left: 50%;
    top: 0;
    transform: translateX(-50%);
    z-index: 2;
    margin: 0;
  }
  .logo img {
    width: 170px;
    height: 170px;
  }
  .navbar {
    min-height: 130px;
  }
  .navbar-collapse {
```

```
    flex-basis: auto;
    flex-grow: 1;
    order: 1;
  }
  .nav-icons {
    order: 2;
    margin-left: 1.5rem;
  }
  .nav-main {
    flex-wrap: nowrap;
    justify-content: space-between;
  }
  .nav-main .nav-link {
    font-size: 0.8rem;
    padding: 2.2rem 0.5rem;
  }
  .logo-gap {
    width: 200px;
  }
}
```

Task 2: Checkout form

New file checkout.html (same head, header and footer as product.html; only the main content below is new). The cart icon in the navbar links to it.

checkout.html (form)

```
<!-- TASK 2 START: HTML5 validation only, no JavaScript -->
    <form class="checkout-form" action="#" method="get">
      <fieldset>
        <legend>Contact</legend>
        <div class="fields">
        <label for="fullname">Full name</label>
        <input type="text" id="fullname" name="fullname" class="form-control" required minlength="3"
          autocomplete="name">
        <span class="error">Please enter your full name.</span>
        <label for="email">Email</label>
        <input type="email" id="email" name="email" class="form-control" required autocomplete="email">
        <span class="error">Enter a valid email address.</span>
        <label for="phone">Phone</label>
        <input type="tel" id="phone" name="phone" class="form-control" placeholder="0300-1234567" required
          pattern="03[0-9]{2}-[0-9]{7}" title="Format: 03XX-XXXXXXX" autocomplete="tel">
        <span class="error">Use the format 03XX-XXXXXXX.</span>
        </div>
      </fieldset>

      <fieldset>
        <legend>Delivery</legend>
        <div class="fields">
        <label for="address">Address</label>
        <input type="text" id="address" name="address" class="form-control" required minlength="5"
          autocomplete="address-line1">
        <span class="error">Please enter your address.</span>
        <label for="city">City</label>
        <select id="city" name="city" class="form-select" required>
          <option value="">Choose a city</option>
          <option>Lahore</option>
          <option>Karachi</option>
          <option>Islamabad</option>
          <option>Rawalpindi</option>
          <option>Faisalabad</option>
          <option>Multan</option>
        </select>
        <span class="error">Please choose a city.</span>
        <label for="delivery">Delivery date</label>
        <input type="date" id="delivery" name="delivery" class="form-control" required min="2026-10-29"
          max="2026-11-29">
        <span class="error">Choose a date from 29 Oct to 29 Nov 2026.</span>
        </div>
      </fieldset>

      <div class="submit-row">
        <button type="submit" class="btn btn-dw btn-lg w-100">Place order</button>
      </div>
    </form>
    <!-- TASK 2 END -->
```

styles.css (Task 2 section)

```
/* ===== TASK 2: checkout form (validation shown only after the user interacts) ===== */
.checkout {
  max-width: 1140px; /* container-fluid keeps the form wide enough at exactly BP */
  padding: 2rem 0.75rem 4rem;
}
.checkout-form {
  display: grid;
  grid-template-columns: 1fr; /* below BP: one column */
  gap: 1.5rem;
  margin-top: 1.5rem;
}
.checkout-form fieldset {
  min-width: 0;
  margin: 0;
  padding: 1rem 0.9rem 1.25rem;
  border: 1px solid #444;
}
.checkout-form legend {
  float: none;
  width: auto;
  padding: 0 0.5rem;
  font-family: "Archivo Black", "Archivo", sans-serif;
  font-size: 1.1rem;
  text-transform: uppercase;
```

```
}
.fields {
  display: grid;
  grid-template-columns: 1fr; /* below BP: label above input */
  gap: 0.35rem;
}
.fields > * {
  min-width: 0; /* lets inputs shrink instead of overflowing the fieldset */
}
.fields label {
  margin-top: 0.6rem;
}
.fields .form-control,
.fields .form-select {
  border-radius: 0;
}
.fields .error {
  display: none; /* hidden until the field is invalid AND touched */
  color: #ff5c5c;
  font-size: 0.85rem;
}
.fields .form-control:user-invalid,
.fields .form-select:user-invalid {
  border: 2px solid #ff3b3b;
}
.fields .form-control:user-invalid + .error,
.fields .form-select:user-invalid + .error {
  display: block;
}

/* From BP and wider: fieldsets side by side, labels in a left column */
@media (min-width: 631px) {
  .checkout-form {
    grid-template-columns: 1fr 1fr;
    align-items: start;
  }
  .submit-row {
    grid-column: 1 / -1; /* button spans both columns */
  }
  .fields {
    grid-template-columns: 5.5rem 1fr;
    align-items: center;
    gap: 0.9rem 0.75rem;
  }
  .fields label {
    margin-top: 0;
  }
  .fields .error {
    grid-column: 2;
    margin-top: -0.5rem;
  }
}
```

Task 3: Cart badge and mini cart

index.html: the cart icon now sits in a wrapper with a badge and the mini-cart panel, and script.js is linked at the end of the body with <script src="script.js"></script>.

index.html (cart icon and mini cart)

```
<!-- TASK 3 START: cart icon with badge and mini cart -->
          <div class="cart-wrap">
            <a href="checkout.html" id="cart-toggle" class="cart-link" aria-label="Cart, 0 items" aria-
              expanded="false" aria-controls="mini-cart">&#128722;<span id="cart-badge" class="cart-
              badge">0</span></a>
            <div id="mini-cart" class="mini-cart">
              <h2 class="mini-cart-title">Your cart</h2>
              <p id="cart-empty">Your cart is empty.</p>
              <ul id="cart-list" class="list-unstyled"></ul>
              <p class="mini-cart-total">Total: <span id="cart-total">$0.00</span></p>
              <a href="checkout.html" class="btn btn-dw w-100">Go to checkout</a>
            </div>
          </div>
          <!-- TASK 3 END -->
```

script.js (new file)

```
// Task 3: cart badge and mini cart (plain JavaScript)

// N = 31, so one product can be added at most (31 % 4) + 2 = 5 times
const MAX_PER_PRODUCT = 5;

// The cart is an array. Each item looks like { name: "Dark Roast Coffee", price: 16.99 }
let cart = [];

const cartToggle = document.getElementById("cart-toggle");
const cartBadge = document.getElementById("cart-badge");
const miniCart = document.getElementById("mini-cart");
const cartList = document.getElementById("cart-list");
const cartEmpty = document.getElementById("cart-empty");
const cartTotal = document.getElementById("cart-total");
const addButtons = document.querySelectorAll(".product-card .btn");

// Read the product name from a card
function getName(button) {
  return button.closest(".product-card").querySelector(".product-name").textContent.trim();
}

// Read the price from a card: "$13.99" becomes 13.99
function getPrice(button) {
  const text = button.closest(".product-card").querySelector(".product-price").textContent;
  return parseFloat(text.replace("$", ""));
}

// Redraw everything that depends on the cart array
function render() {
  // 1. badge
  cartBadge.textContent = cart.length;
  cartToggle.setAttribute("aria-label", "Cart, " + cart.length + " items");

  // 2. list of items with a Remove button
  cartList.innerHTML = "";
  cart.forEach(function (item, index) {
    const li = document.createElement("li");

    const name = document.createElement("span");
    name.className = "item-name";
    name.textContent = item.name;

    const price = document.createElement("span");
    price.textContent = "$" + item.price.toFixed(2);

    const remove = document.createElement("button");
    remove.type = "button";
    remove.className = "btn btn-sm btn-outline-light";
    remove.textContent = "Remove";
    remove.dataset.index = index;

    li.append(name, price, remove);
    cartList.append(li);
  });

  // 3. empty message
  cartEmpty.hidden = cart.length > 0;

  // 4. total, calculated with reduce()
```

```
  const total = cart.reduce(function (sum, item) {
    return sum + item.price;
  }, 0);
  cartTotal.textContent = "$" + total.toFixed(2);

  // 5. limit per product: disable the button when it has been added 5 times
  addButtons.forEach(function (button) {
    const name = getName(button);
    const count = cart.filter(function (item) {
      return item.name === name;
    }).length;

    if (count >= MAX_PER_PRODUCT) {
      button.textContent = "Limit reached";
      button.disabled = true;
    } else {
      button.textContent = "Add to Cart";
      button.disabled = false;
    }
  });
}

// Add to Cart buttons
addButtons.forEach(function (button) {
  button.addEventListener("click", function () {
    cart.push({ name: getName(button), price: getPrice(button) });
    render();
  });
});

// Remove buttons (one listener on the list handles all of them)
cartList.addEventListener("click", function (event) {
  if (event.target.matches("button")) {
    cart.splice(Number(event.target.dataset.index), 1);
    render();
  }
});

// Cart icon opens and closes the mini cart
cartToggle.addEventListener("click", function (event) {
  event.preventDefault(); // stay on the page instead of going to checkout.html
  const isOpen = miniCart.classList.toggle("open");
  cartToggle.setAttribute("aria-expanded", isOpen);
});

render();
```

styles.css (Task 3 section)

```
/* ===== TASK 3: cart badge and mini cart ===== */
.cart-wrap {
  position: relative;
}
.cart-link {
  position: relative;
  display: inline-block;
}
.cart-badge {
  position: absolute;
  top: -8px;
  right: -12px;
  min-width: 1.25rem;
  padding: 0 0.3rem;
  border-radius: 0.65rem;
  background: var(--red);
  color: #fff;
  font-family: system-ui, sans-serif;
  font-size: 0.75rem;
  line-height: 1.25rem;
  text-align: center;
}
.mini-cart {
  display: none;
  position: absolute;
  top: 100%;
  right: 0;
  z-index: 30;
  width: min(320px, 90vw);
  padding: 1rem;
  background: #111;
  border-top: 4px solid var(--accent);
  box-shadow: 0 8px 24px rgba(0, 0, 0, 0.6);
  font-family: "Archivo", system-ui, sans-serif;
  font-size: 0.95rem;
```

```
}
.mini-cart.open {
  display: block;
}
.mini-cart-title {
  font-size: 1rem;
}
.mini-cart li {
  display: flex;
  align-items: center;
  gap: 0.5rem;
  padding: 0.4rem 0;
  border-bottom: 1px solid #333;
}
.mini-cart .item-name {
  flex: 1;
}
.mini-cart-total {
  margin: 0.75rem 0;
  font-weight: 600;
}
```